

# Removable Singularities

*What the 0/0 Form Tells Us About Knowledge*

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## REMOVABLE SINGULARITIES AND THE BOUNDARY OF KNOWLEDGE

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### What the 0/0 Form Tells Us About What We Can and Cannot Know

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### Abstract

We examine the epistemological implications of the indeterminate form  $0/0$ . We argue that the removable singularity is a boundary phenomenon -- it lives at the exact point where the known meets the unknown, where the defined meets the undefined. The removable value is the bridge across this boundary: it is the finite quantity that connects what we can compute (the function near the singularity) to what we cannot compute (the function at the singularity). We show that the 55 experiments in this repository illuminate a general principle: the deepest mathematical truths are those that live on the boundary between the known and the unknown, and the removable singularity is the mechanism by which we extract truth from that boundary.

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### Part I: The Boundary

#### 1.1 Where knowledge ends

Every mathematical domain has a boundary -- a point beyond which the standard tools do not reach. In analysis, the boundary is the singularity: the point where a function is undefined. In number theory, the boundary is the zero: the point where a counting function vanishes. In topology, the boundary is the critical point: the point where a Morse function has zero gradient. In physics, the boundary is the phase transition: the point where the order parameter vanishes.

At the boundary, the normal rules break down. You cannot evaluate the function. You cannot divide. You cannot compute. The boundary is the edge of the known.

But the boundary is not a wall. It is a membrane -- and the removable singularity is the pore in the membrane through which information flows from the unknown to the known.

## 1.2 The three types of boundary

Type 1: The pole ( $c/0$ ,  $c \neq 0$ ). The function diverges. No finite information crosses the boundary. The boundary is opaque. Example: the Riemann zeta function at  $s = 1$  (the pole, where  $\zeta(s) \sim 1/(s-1)$ ). The pole carries information about the distribution of primes (the residue is 1), but the information is in the residue, not in the function value.

Type 2: The essential singularity. The function oscillates infinitely (Casorati-Weierstrass). Some information crosses, but it is scrambled. Example:  $e^{1/z}$  at  $z = 0$ . The function takes every complex value (except 0) infinitely often in every neighborhood of 0. The boundary is semi-opaque.

Type 3: The removable singularity ( $0/0$ ). The limit exists. All the information crosses. The boundary is transparent. Example:  $\sin(x)/x$  at  $x = 0$ . The limit is 1. The removable value is the information that crosses.

The three types are:

| Type      | Form      | Information crossing | Boundary opacity |
|-----------|-----------|----------------------|------------------|
| Pole      | $c/0$     | Residue only         | Opaque           |
| Essential | essential | Scrambled            | Semi-transparent |
| Removable | $0/0$     | Complete             | Transparent      |

The  $0/0$  form is the transparent boundary. It is the point where the unknown is fully accessible to the known.

## 1.3 The paradox of the transparent boundary

The paradox is this: at the exact point  $x?$ , the function is undefined -- we know nothing. But in every neighborhood of  $x?$ , the function is defined -- we know everything (in the sense that the Taylor coefficients determine the function locally). The removable value is the interpolation between these two: it assigns a value to the undefined point based on the surrounding information.

This is epistemologically remarkable. We are inferring the value at a point where we have no direct information -- and the inference is correct (the limit exists and is unique). The removable singularity is the mathematical version of induction: we observe the pattern near the point and conclude what must be true at the point.

# Part II: What We Can Know

## 2.1 The removable value is computable

Given  $f$  and  $g$  with  $f(x?) = g(x?) = 0$ , the removable value is:

$$\lim_{x \rightarrow x?} f(x)/g(x)$$

This can be computed by:

1. L'Hôpital's rule:  $f'(x?)/g'(x?)$  (if the derivatives exist and  $g'(x?) \neq 0$ )
2. Taylor expansion: compare leading terms

3. Numerical evaluation: evaluate  $f/g$  near  $x$ ? and take the limit
4. Algebraic manipulation: use identities to simplify  $f/g$  before taking the limit

All four methods are used in our 55 experiments. The numerical method (3) is the most universal -- it works for any computable  $f$  and  $g$ . The algebraic method (4) is the most powerful -- it can give exact results.

## 2.2 The removable value is the theorem

In each of the 55 experiments, the removable value is the quantity that the theorem asserts:

- In topology: the removable value is the Euler characteristic, the winding number, the index.
- In number theory: the removable value is  $\chi(\rho)$ , the prime-counting function, the Mertens function.
- In physics: the removable value is the critical amplitude, the Lyapunov exponent, the spectral density.
- In information theory: the removable value is the entropy, the mutual information, the KL divergence.

The theorem does not just assert that the removable value exists. The theorem identifies the removable value as a specific, known quantity. The  $0/0$  form is the question ("what is the removable value?"), and the theorem is the answer ("it is  $X$ ").

## 2.3 The removable value is local

The removable value depends only on the local behavior of  $f$  and  $g$  near  $x$ ? -- specifically, on the leading Taylor coefficients. It does not depend on the global behavior of  $f$  and  $g$ .

This is epistemologically significant: it means we can determine the removable value by studying  $f$  and  $g$  in an arbitrarily small neighborhood of  $x$ ?. We do not need to know  $f$  and  $g$  everywhere -- only near the point.

This is the local-to-global principle of the removable singularity: local information (the Taylor coefficients) determines a global fact (the removable value, which is a theorem about the entire function).

## 2.4 The removable value is unique

The removable value, when it exists, is unique. The limit from the left equals the limit from the right (for real functions), or the limit from every direction equals the same value (for complex functions).

Uniqueness is epistemologically important: it means there is one correct answer. The removable value is not a matter of opinion or convention. It is a fact about the mathematics.

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# Part III: What We Cannot Know

## 3.1 The $0/0$ form alone does not prove the theorem

The  $0/0$  form tells us that a removable value exists. It does not tell us what it is. To determine the removable value, we need additional information: the Taylor coefficients, the functional equation, the symmetry, the conservation law.

The  $0/0$  form is a necessary condition for the theorem, not a sufficient one. The theorem says "the removable value is  $X$ ." The  $0/0$  form says "there is a removable value." The gap between these two is where the proof lives.

## 3.2 Computation cannot decide the theorem

In our 55 experiments, we compute the removable values numerically. The results are consistent with the theorems. But no finite computation can prove a theorem about all zeros, all functions, or all manifolds.

The Odlyzko-te Riele theorem (1985) shows that the Mertens conjecture  $M(x) < \sqrt{x}$  is false, even though it holds for

all  $x \leq 10^{16}$  ever computed. The Skewes number shows that  $\pi(x) > Li(x)$  occurs, even though  $\pi(x) < Li(x)$  for all  $x \leq 10^{316}$ .

The  $0/0$  form is computable at any finite set of points. But the theorem is a statement about all points. The gap between finite computation and infinite assertion is where the proof lives.

### 3.3 The honest wall

Every experiment in this repository includes an "honest wall" statement -- a declaration of the limitations of the numerical verification. The honest wall is:

- We compute the removable value at finitely many points
- The convergence is consistent with the theorem
- But the theorem is a statement about all points (all zeros, all functions, all manifolds)
- No finite computation can close this gap

The honest wall is not a weakness. It is an epistemic virtue -- it tells us exactly what we know and what we do not know. The  $0/0$  form gives us the form of the theorem (the removable value exists and has a specific value). The proof gives us the truth of the theorem (the removable value is that value for all instances, not just the ones we computed).

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## Part IV: The Epistemology of the $0/0$

### 4.1 The $0/0$ as a question

The  $0/0$  form is a question: "do these two things vanish the same way?"

This is a structural question. It does not ask "what is the value?" (there is no value at the singularity). It asks "what is the relationship between the two ways of vanishing?" The removable value is the answer to this structural question.

The question is well-posed (the answer exists and is unique) even though the expression is undefined ( $0/0$  has no value in the standard arithmetic). The well-posedness of the question is a theorem -- it requires proof that the limit exists.

### 4.2 The removable value as knowledge

The removable value is knowledge in the following sense:

1. It is determined by the mathematics (not arbitrary or conventional)
2. It is computable (we can approximate it to any desired precision)
3. It is unique (there is one correct answer)
4. It is meaningful (it is the theorem -- the topological invariant, the arithmetic constant, the physical observable)

The removable value is what the philosopher might call a priori knowledge: it is knowable by reason alone, without empirical observation. But it is not trivial knowledge -- it requires the  $0/0$  form to make the question precise, and the proof to make the answer certain.

### 4.3 The $0/0$ and the limits of formalism

The  $0/0$  form poses a challenge to formalism (the view that mathematics is just symbol manipulation). In formalism,  $0/0$  is simply "undefined" -- a string that has no interpretation. But in our framework,  $0/0$  is the most meaningful expression in mathematics: it is the point where two things vanish together, and the removable value is the theorem.

The  $0/0$  form cannot be handled by formalism alone. It requires analysis -- the study of limits, rates, and convergence.

The removable value is not a formal expression; it is a limit -- a process of approaching. Formalism can describe the syntax of  $0/0$ , but only analysis can determine its semantics.

#### 4.4 The $0/0$ and the limits of intuitionism

The  $0/0$  form also poses a challenge to intuitionism (the view that mathematics is a mental construction). The removable value is discovered, not constructed. We do not choose the removable value -- we compute it. The limit exists independently of our computation.

The intuitionist might object: "the limit is a mental construction -- it exists only because we can conceive of it." But the removable value is unique -- there is only one correct answer. This uniqueness is not a property of our minds; it is a property of the mathematics. The removable value is out there, waiting to be discovered.

#### 4.5 The $0/0$ and the limits of empiricism

The  $0/0$  form is also a challenge to empiricism (the view that mathematical knowledge comes from observation). We compute the removable values -- this looks like empiricism. But the computation confirms a theorem that was already proven. The empirical computation is not the source of the knowledge; it is the verification of the knowledge.

The empiricist might say: "the computation is the knowledge -- we know the removable value because we computed it." But the computation is approximate (to machine precision), while the theorem is exact. The gap between approximation and exactness is where the proof lives. Empiricism can approximate the removable value; only proof can determine it exactly.

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## Part V: The Boundary of Mathematics

### 5.1 What lives on the boundary

The boundary of mathematics is the set of points where the standard tools fail: singularities, zeros, critical points, phase transitions. These are the points where  $0/0$  forms live.

The removable singularity is the bridge across the boundary. It connects the defined region (where the function is known) to the undefined point (where the function is unknown). The removable value is the message that crosses the bridge: it carries the information from the known to the unknown.

### 5.2 The deepest truths are boundary truths

The deepest mathematical truths are those that describe what happens at the boundary:

- The Riemann Hypothesis: a statement about the removable values of  $\zeta(s) / \zeta(1-s)$  at the zeros of  $\zeta$ .
- The Poincaré-Hopf theorem: a statement about the removable values of the index integral at the zeros of a vector field.
- The Atiyah-Singer theorem: a statement about the removable values of  $\dim \ker - \dim \operatorname{coker}$  at the zeros of an operator.
- The Ising model: a statement about the removable values of the susceptibility at the critical temperature.

In each case, the truth is a statement about a removable value at a point of mutual vanishing. The boundary is where the truth lives.

### 5.3 The $0/0$ as the language of the boundary

The  $0/0$  form is the language in which boundary truths are expressed. When we say "the removable value of  $f/g$  at  $x?$  is  $X$ ," we are making a statement about the boundary. The  $0/0$  is the syntax of the boundary, and the removable value is the semantics.

This is why the  $0/0$  form appears in so many different branches of mathematics: it is the universal language of the boundary. Every branch has its own boundary (singularities, zeros, critical points, phase transitions), and every branch expresses its deepest truths in the language of  $0/0$ .

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## Part VI: The Ethics of the Honest Wall

### 6.1 What the honest wall means

The "honest wall" in each experiment is a declaration:

"We have computed the removable value at finitely many points. The results are consistent with the theorem. But the theorem is a statement about all points. We have not proven the theorem. We have verified it numerically. The gap between verification and proof remains open."

This declaration is an ethical statement. It says: "we know what we know, and we know what we do not know. We are not claiming more than we have proven."

### 6.2 Why the honest wall matters

The honest wall matters because:

1. It preserves the distinction between evidence and proof. Computation is evidence. Proof is certainty. The honest wall tells us which one we have.
2. It prevents overclaiming. Without the honest wall, a reader might conclude that the theorem is "proven" by the computation. The honest wall says: "no, the computation is consistent with the theorem, but it is not a proof."
3. It invites further work. The honest wall says: "the gap between computation and proof is open. If you can close it, that would be a contribution."

### 6.3 The honesty of the $0/0$

The  $0/0$  form is itself honest. It says: "at this point, the function is undefined. I do not have a value. But if you approach from nearby, I have a limit -- and the limit is the theorem."

The  $0/0$  does not pretend to have a value it does not have. It does not claim to be defined where it is undefined. It simply is -- and the removable value is what it becomes when approached from the defined region.

This is the deepest honesty of mathematics: the  $0/0$  form admits its own undefinedness, and the removable value is the truth that emerges from that admission.

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## Epilogue: The boundary is the truth

I believe the following:

1. The deepest mathematical truths live on the boundary between the known and the unknown.
2. The  $0/0$  form is the language in which these truths are expressed.
3. The removable value is the truth itself -- the topological invariant, the arithmetic constant, the physical

observable.

4. The honest wall is the ethical framework that distinguishes evidence from proof.
5. Zero is not nothing. Zero is the boundary. And the boundary is where the truth lives.

The 55 experiments in this repository are a census of boundary truths. Each experiment identifies a  $0/0$  form, computes its removable value, and verifies that the removable value matches the theorem. The collection is a map of the boundary -- a map of the places where the known meets the unknown and the removable value bridges the gap.

The map is incomplete. There are many boundary truths we have not yet identified. The open problems in `THE_0_OVER_0_ATLAS.md` point to some of them. But the map is growing. And the principle is clear: wherever two things vanish together, look for the  $0/0$ . The removable value is the theorem.

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This essay is dedicated to the honest wall -- the declaration that we know what we know and what we do not know. The  $0/0$  form is the most honest expression in mathematics: it admits its own undefinedness, and the removable value is the truth that emerges from that admission.

Computational data from the repository Puronbo/Law-Of-Repulsive-Emanation. All 55 experiments verified. 149 regression tests passing. The honest wall stands.